

Database Engine: SQLite 3 via RESTful Express API

3.1. Public Promotional Web Portal

PUBLIC & GUEST ACCESS

A lightweight, high-performance web portal providing open public access to the catalog and serving as the primary distribution hub for the mobile client.

- **Zero-Login Live Catalog Search:** Allows external visitors and students to perform real-time searches across the complete book collection by Title, Author, Category, or ISBN without requiring login credentials.
- **Live Inventory Stock Indicators:** Displays physical shelf availability (e.g., total copies vs. current available count) synchronized dynamically with the main library database.
- **Direct APK Distribution & Sideloadable Guide:** Hosts the official Android Application Package (.apk) direct download link alongside an interactive, step-by-step installation walkthrough for granting device permissions.
- **System Synchronization Visualizer:** Provides visual conceptualization of real-time multi-platform data flow between mobile clients, the web portal, and the librarian administrative database.
- **Institutional FAQ & Policies Knowledge Base:** An interactive accordion module providing rapid answers on borrowing rules, loan durations, account verification, and penalty guidelines.
- **Responsive High-Performance UI:** Implements modern CSS grid/flexbox layouts and micro-interactions optimized for rapid rendering across mobile and desktop browsers without third-party framework overhead.

3.2. Web-Based Librarian Desktop Management Portal

LIBRARIANS & STAFF

A secure Single-Page Application (SPA) tailored for circulation desk operations, inventory control, loan transactions, and operational reporting.

- **Role-Based Access Control (RBAC) & Cryptographic Security:** Employs PBKDF2 SHA-512 password hashing with 16-byte random salts, timing-safe equality verification, and 24-hour signed JWT session tokens. Distinguishes administrative rights between Head Librarians and staff members.
- **Comprehensive Book Inventory Management:** Full CRUD capabilities for titles, unique ISBN validation, category assignment, synopsis storage, and automated available-stock recalculation upon copy adjustments.
- **Active Loan & Circulation Processing:** Facilitates checkout operations with student record linkage and automated due date assignment; processes book returns with physical condition tagging (Good, Damaged, Lost).
- **Automated Overdue & Dynamic Penalty Engine:** Compares active transaction due dates against current system timestamps on demand, transitioning expired loans to 'overdue' status and computing penalty fines dynamically.
- **Student Record & Reservation Queue Administration:** Provides a centralized directory of enrolled student profiles and an active reservation management console to accept, queue, or fulfill hold requests.
- **Analytical Reporting & Data Visualization:** Generates real-time visual summaries using graphical charts (e.g., circulation volume, top borrowed categories, overdue percentages, and inventory ratios).

3.3. Student Mobile Application (Android Client)

ENROLLED STUDENTS

A dedicated mobile client enabling students to engage with library services 24/7 beyond physical campus network boundaries.

- **Student Authentication & Digital Library Identity:** Secure credential management linking student profiles to active institutional records, displaying active borrow limits and account status.
- **Categorized Book Exploration & Filtering:** Fast category-based browsing (e.g., Technology, History, Mathematics, Literature) with real-time text query filtering and detailed book metadata views.
- **Remote Book Reservation System:** Enables students to submit remote hold requests for available shelf titles, tracking reservation progress through Pending, Ready for Pickup, and Fulfilled lifecycles.
- **Active Borrowing & Due Date Countdown Dashboard:** Visual tracking of currently borrowed books featuring color-coded status badges and automated countdowns to prevent late returns.
- **Penalty & Fine Transparency:** Displays prospective penalty charges for overdue items to maintain complete financial and accountability transparency prior to physical return.